

Aakash Shah

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EDUCATION

University of North Carolina at Charlotte, Charlotte, NC

May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures and Algorithms, Computer Networks, Computer Systems, Computer Architecture

EXPERIENCE

Microsoft, Redmond, WA

June 2026 - August 2026

Software Engineer Intern

- Built **Python** and **Azure** Functions collectors centralizing failure telemetry across **200+** production data pipelines
- Developed **PyTorch** and **LightGBM** models on Azure forecasting incidents 24 hours ahead with **94%** precision
- Clustered recurring pipeline failures using **OpenAI** embeddings, UMAP, and HDBSCAN to automate discovery
- Optimized embeddings with versioned caching and deduplication modeling, cutting repeat latency by **88%**
- Engineered **FastAPI** services and **Grafana** dashboards to surface pipeline health, reducing triage time by **35%**

SAS Institute, Cary, NC

May 2025 - May 2026

Software Engineer Intern

- Built Machine Learning pipelines for anomaly detection in industrial IoT, cutting model training time by **50%**
- Designed real-time workflows for HVAC and solar data, reducing false alert rates by **40%** in edge systems
- Developed **Go** and **Python** APIs to stream predictions, lowering client response latency by **30%**
- Improved prediction workflow reliability by validating sensor data and standardizing model output formats

AI4Health at UNC Charlotte, Charlotte, NC

March 2026 - August 2026

Undergraduate Researcher

- Architected an Android biometric engine in **Kotlin** and **Jetpack Compose**, creating multimodal identity layer
- Fused touch dynamics and heart rate data into a unified behavioral profile for passive authentication real time
- Trained on-device **PyTorch** ML models to detect anomalous user behavior with sub-**100ms** inference latency
- Led **DARPA** affiliated mobile security research, shipping real time verification pipelines via on device interface

VisioniOS Lab at UNC Charlotte, Charlotte, NC

August 2025 - November 2025

Undergraduate Researcher

- Engineered a **CoreML** detection engine on Vision Pro using **Swift**, achieving sub-**80ms** latency and **92%** accuracy
- Built a spatial audio cue system with RealityKit for visually impaired users, reducing navigation errors by **38%**
- Developed a dynamic spatial awareness pipeline with **ARKit** in visionOS, improving object detection by **45%**

New England Investment Consulting Group (NEW EIC), Remote

January 2025 - April 2025

Software Engineer Intern

- Built internal systems with **MongoDB**, **TypeScript**, and **Docker** to streamline workflows for **50+** developers
- Developed automated issue tracking and webhook-based Slack notifications, reducing response time by **45%**

PROJECTS

Gravity Terminal | Go, TypeScript, React, PostgreSQL, Solana, WebSockets

- Built two leg arbitrage engine in **Go** across Kalshi and Polymarket with slippage limits and automated hedge retry
- Designed TTL-aware cache layer across **10+** live feeds (Binance, GDELT, ESPN) cutting redundant calls and latency
- Routed **\$50k+** in on-chain Solana transactions with priority fee optimization and LLM research engine for analysis

Gold Mine UNCC | Swift, Kotlin, JetPack Compose, MapKit, SwiftSoup, Core Data

- Developed iOS and Android apps used by **3,500** students, providing transit, maps, and campus services
- Implemented concurrent API fetching, SwiftSoup parsing, and Core Data caching for offline access
- Built Jetpack Compose Android client with MVVM architecture, adaptive Material Design, and Room persistence

TECHNICAL SKILLS

Languages: Python, Go, Swift, C++, C, TypeScript, Kotlin, Java, JavaScript, Angular, React-Native, HTML/CSS

Frameworks/Libraries: React, Colly, FastAPI, Flask, Next.JS, GraphQL, Prisma, Supabase, Express, PostgreSQL, AWS

Concepts: AI Infrastructure, Big Data, Data Lakes, Machine Learning, Databases, Full Stack, Backend, Mobile, NLP